WSPRadar.org reconstruction & comparison

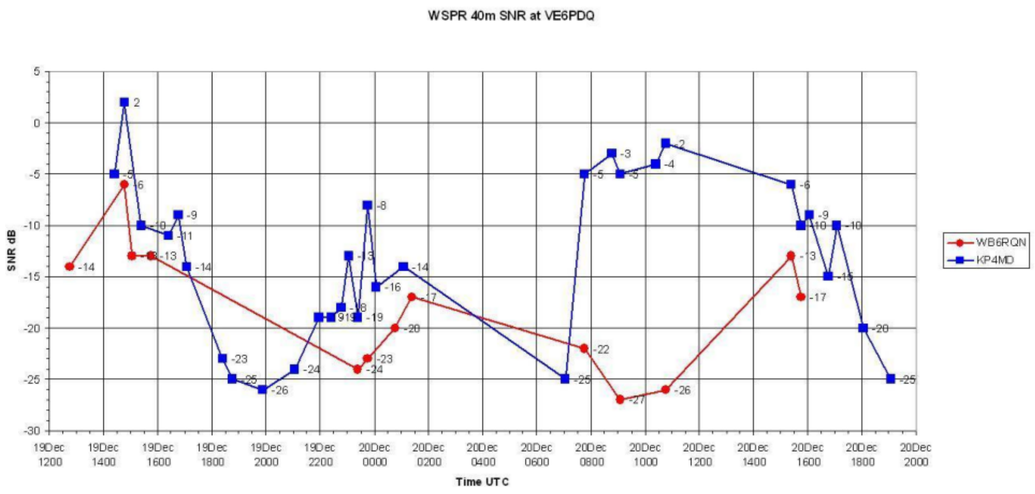
Reconciled RX direction: VE6PDQ transmitting to KP4MD / WB6RQN

Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

Source figure: **Figure 6** · Amateur-radio technical article and club presentation · 2011

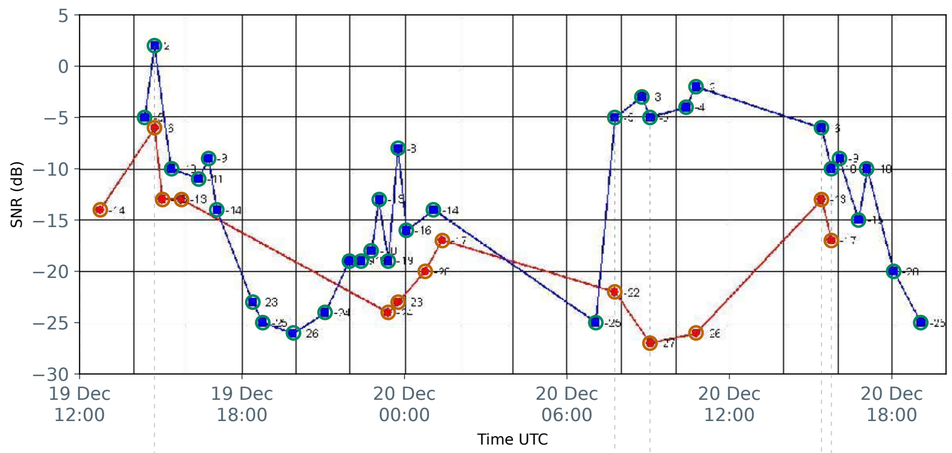
Demo: Historical RX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | VE6PDQ (DO34 / DO34IR) to KP4MD / WB6RQN | Reference correction 0 dB

Panel A - Original image from publication



Archive fields identify VE6PDQ → KP4MD / WB6RQN, opposite to the original direction label. The published image is retained unchanged.

Panel B — Archived reports on the published figure



44 archived reports; five spot pairs

KP4MD is Target; WB6RQN is Reference.

Panel B overlays all 44 archived reports on the retained publication plot. The reconstruction identifies 44 published-point matches with agreement at the printed whole-dB SNR resolution. Archive timestamps are unchanged; only the stated power normalization is applied. Original lines supply no intermediate measurements.

A spot pair requires both receivers to report the same remote callsign and full locator in the same two-minute WSPR cycle. Three-hour bins group the display only.

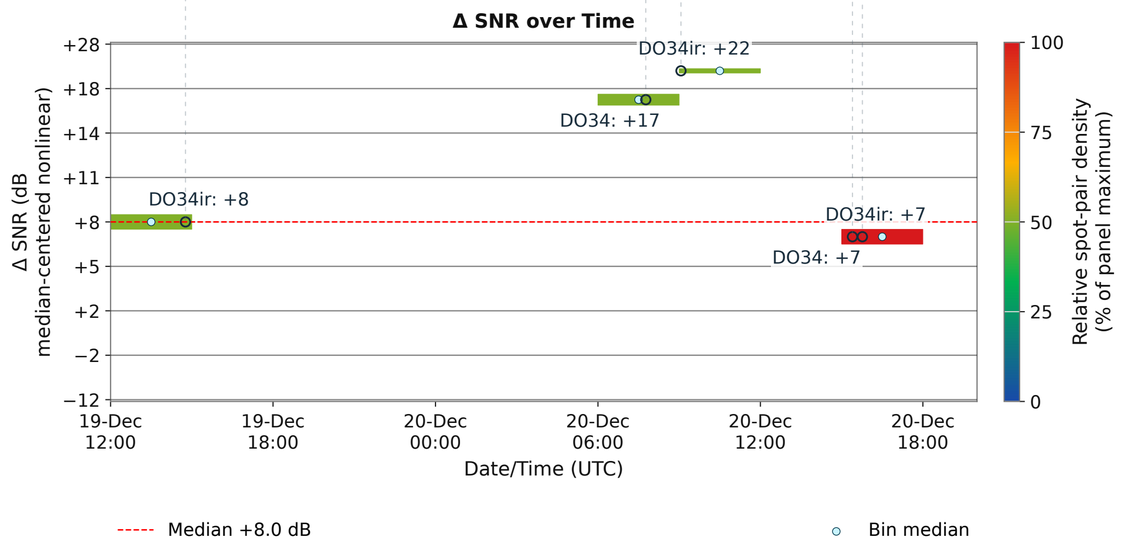
All five pairs favor KP4MD: +7 to +22 dB, with a pooled median of +8 dB. Three pairs belong to DO34IR and two to DO34. These five observations do not establish a constant advantage.

VE6PDQ reported 37 dBm (approximately 5 W). Panel B shows raw SNR; normalization to 30 dBm lowers both receivers' values by 7 dB, leaving ΔSNR unchanged.

DO34IR lies within DO34. Their different reported precision does not establish movement; the identities remain separate when pairing.

Panel C - WSPRadar paired ΔSNR

5 spot pairs | KP4MD - WB6RQN | exact full locator identity



Circles: exact pairs; small markers: three-hour bin medians. Colour shows relative density, not confidence. The horizontal dashed line is the paired median. Read the nonlinear axis by its dB labels.

WSPRadar settings: peer distance < 5 000 km; minimum spot pairs per station: 1; all directions; solar state: all; special-callsign and moving-station exclusions: off.

WSPRadar.org by Dr. Markus Brosch (DL1MKS)

WSPRadar.org reconstruction & comparison

Observed RX coverage over time: counts and ΔDecode

Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

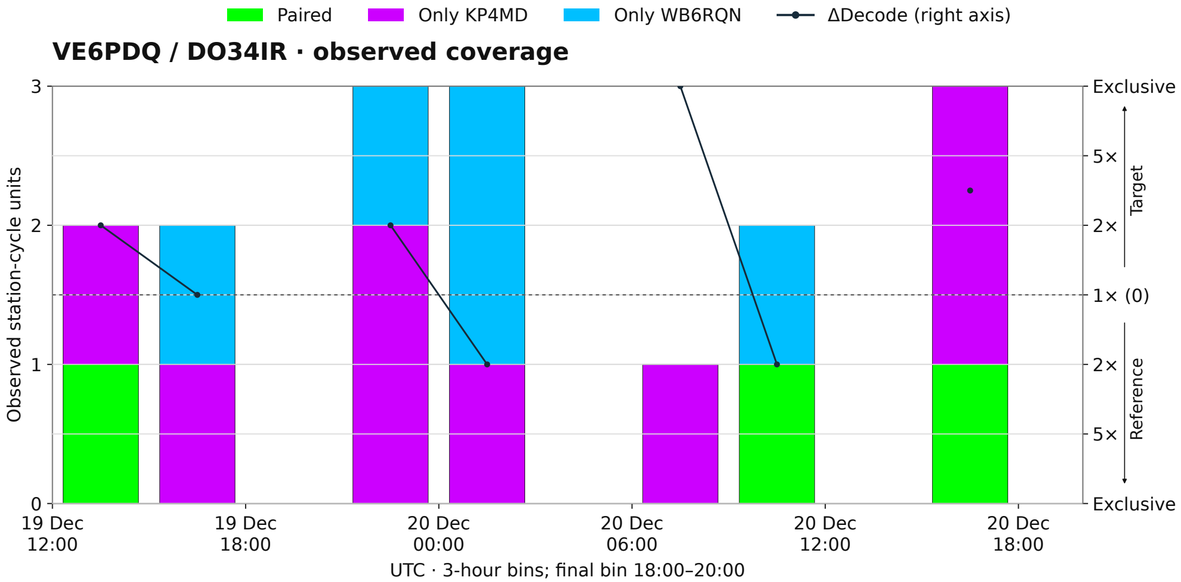
Source figure: **Figure 6** · Amateur-radio technical article and club presentation · 2011

Demo: Historical RX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | VE6PDQ (DO34 / DO34IR) to KP4MD / WB6RQN | Reference correction 0 dB

Overview · same-cycle, full-locator outcomes

Target: KP4MD; Reference: WB6RQN.

Selected identity	Only T	Spot pairs	Only R	Station-cycles	ΔDecode (×)
VE6PDQ / DO34IR	8	3	5	16	1.38×
VE6PDQ / DO34	18	2	3	23	4.00×
Selected path total	26	5	8	39	2.38×

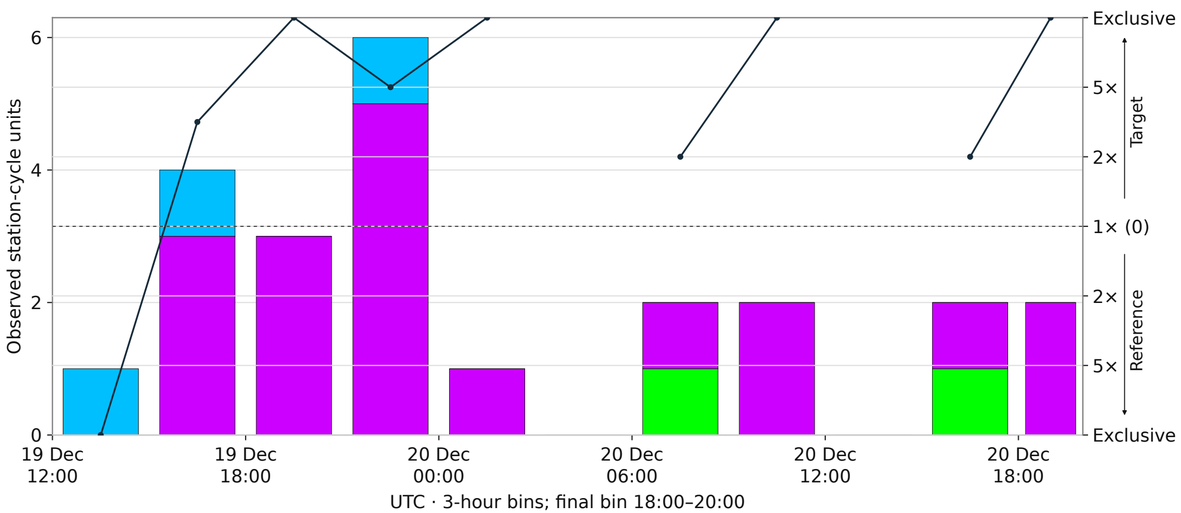


Station-cycle: one observed remote callsign/locator/cycle event; timestamps may repeat.
Bars and outcome columns count events; fold factors compare report totals.

VE6PDQ / DO34IR · exact bin counts

UTC bin	Only T	Spot pairs	Only R	ΔDecode (×)
19 Dec 12:00–15:00	1	1	0	2.00×
19 Dec 15:00–18:00	1	0	1	1.00×
19 Dec 18:00–21:00	0	0	0	
19 Dec 21:00–00:00	2	0	1	2.00×
20 Dec 00:00–03:00	1	0	2	0.50×
20 Dec 03:00–06:00	0	0	0	
20 Dec 06:00–09:00	1	0	0	∞×
20 Dec 09:00–12:00	0	1	1	0.50×
20 Dec 12:00–15:00	0	0	0	
20 Dec 15:00–18:00	2	1	0	3.00×
20 Dec 18:00–20:00	0	0	0	

VE6PDQ / DO34 · observed coverage



VE6PDQ / DO34 · exact bin counts

UTC bin	Only T	Spot pairs	Only R	ΔDecode (×)
19 Dec 12:00–15:00	0	0	1	0.00×
19 Dec 15:00–18:00	3	0	1	3.00×
19 Dec 18:00–21:00	3	0	0	∞×
19 Dec 21:00–00:00	5	0	1	5.00×
20 Dec 00:00–03:00	1	0	0	∞×
20 Dec 03:00–06:00	0	0	0	
20 Dec 06:00–09:00	1	1	0	2.00×
20 Dec 09:00–12:00	2	0	0	∞×
20 Dec 12:00–15:00	0	0	0	
20 Dec 15:00–18:00	1	1	0	2.00×
20 Dec 18:00–20:00	2	0	0	∞×

These 44 reports yield 31 at KP4MD and 13 at WB6RQN: 2.38×. They represent 39 observed station-cycles; each spot pair contributes two reports. Differing operating schedules limit interpretation: this is an observed count ratio, not isolated antenna gain or an absolute decoding probability. Table factor = (Only T + Spot pairs)/(Only R + Spot pairs). 1×: equal; 0.5×: Reference 2×; ∞×: Target only; 0×: Reference only. Blank: no observations; transmission and decoding status unknown. Normalization cannot recover absent reports or equalize operating opportunities.

WSPRadar settings: peer distance < 5 000 km; minimum spot pairs per station: 1; all directions; solar state: all; special-callsign and moving-station exclusions: off.